

Domain Concept and Database Integration

Domain Modelling

What problem does the system solve?

- Incidents come from multiple platforms, making reporting, tracking, and coordination difficult.
- Incident severity is declared manually and can be inconsistent.
- Incident management is decentralized, resulting in fragmented information and manual effort.
- Handling incidents and creating tickets for developers requires significant manual work.

Who is it for?

- Clients who report incidents
- Developers who handle and resolve incidents
- Team leads who monitor incidents and sometimes assign them manually

Domain Concept

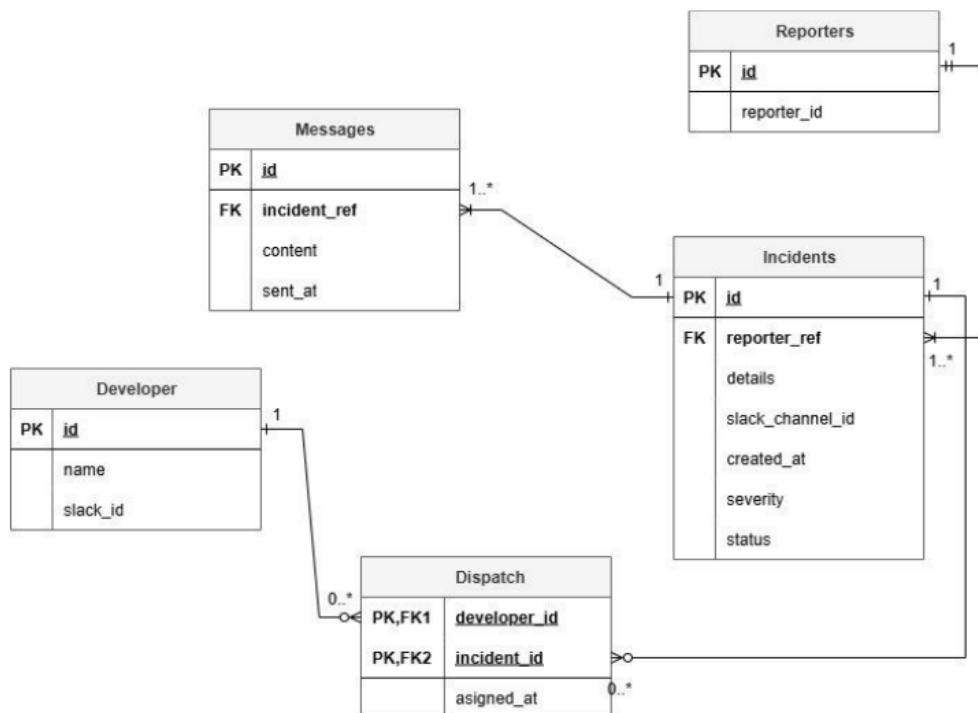
Term	Meaning	Domain Objects
Incident	A reported issue that requires investigation and resolution	Aggregate root
Platform	A source or channel from which an incident is reported.	Value object
Severity	The business impact level of an incident.	Value object
Status	The lifecycle state of an incident	Value object
Reporter	The party that reported the incident	Value object
Developer	A person responsible for handling and resolving incidents.	External reference (DeveloperId)
Dispatch	The act of sending an incident to developers for handling	Domain behavior
Assignment	The automated or manual allocation of an incident by a team lead	Domain behavior
Assignee	The developer responsible for handling an incident	Value object (DeveloperId)

Database PostgreSQL

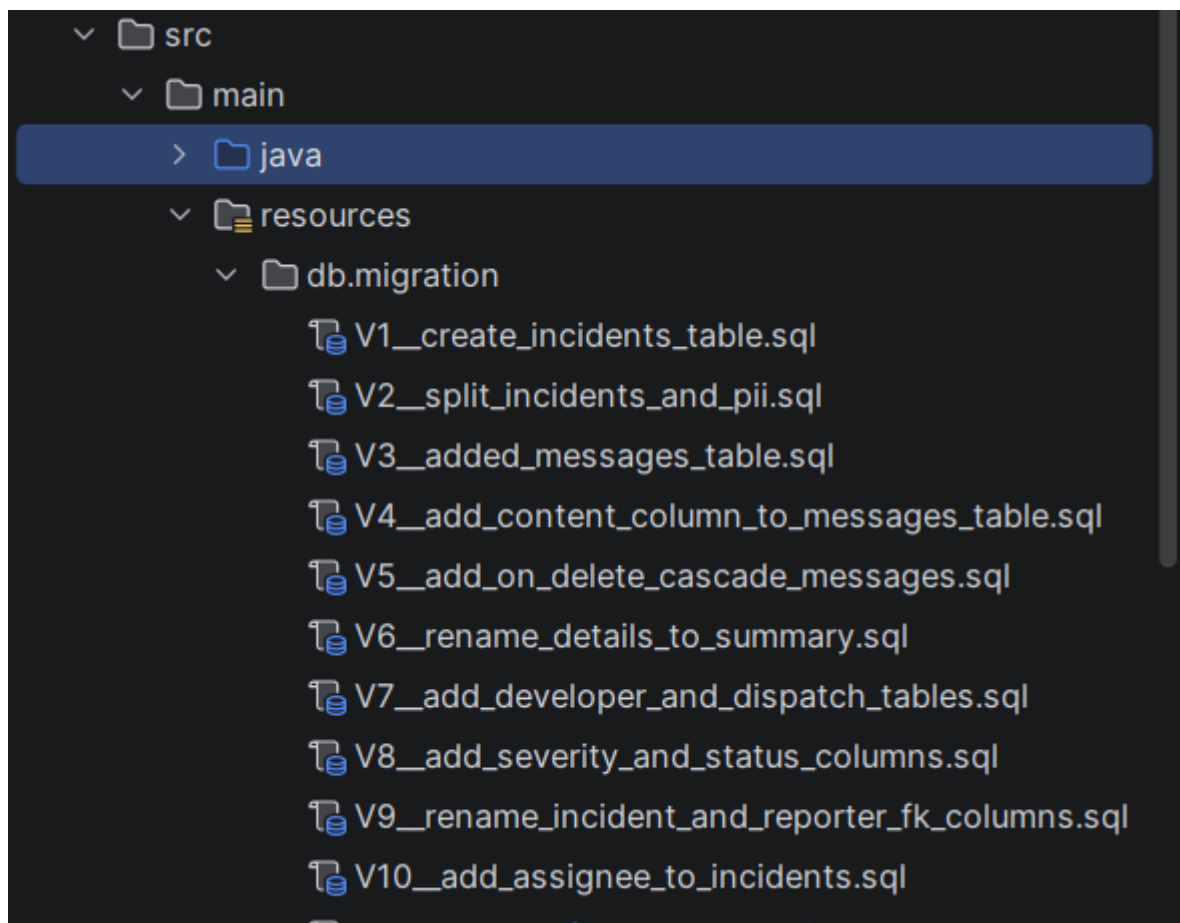
Database schema design

Requirements

1. A reporter must be associated with at least one reported incident.
2. A reporter can send multiple messages, which may lead to multiple incident tickets being created.
3. An incident consists of multiple messages forming a conversation but always requires at least one initial message.
4. Developers can exist in the system without being assigned to any incident.
5. One incident can have multiple developers assigned to it.



Flyway Database Migrations



Shortcuts for running the Database

Requirements

- Docker
- Bash

Run the following command:

```
docker pull postgres:16.  
docker run --name incident_postgres \  
-e POSTGRES_USER=goup_user \  
-e POSTGRES_PASSWORD=group_pass \  
-e POSTGRES_DB=incident_db \  
-p 5432:5432 \  
-d postgres:16
```

Pull the PostgreSQL 16 Docker image and run it as a container named `incident_postgres`. During startup, it automatically creates the `incident_db` database using the configured username and password.

Variables

`POSTGRES_DB=incident_db`

`POSTGRES_USER=<your-username>`

`POSTGRES_PASSWORD=<your-password>`